

CDFD Runtime Paper 08: Proto-Regime, Boundary Conditions, and Theoretical Discovery

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Abstract

Executive synthesis. The Proto-Regime is best understood as a boundary-condition discovery candidate: a high-flow, low-constraint state in which ordinary interpretation of Ψ_s breaks down. Earlier drafts described it too strongly. This upgraded paper frames the Proto-Regime as a reproducible runtime stress test and a theoretical hypothesis, not as proof of faster-than- light travel, pre-Big-Bang physics, or deployable vacuum engineering.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or re-distribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η , λ , μ , ν , ρ , σ , τ , and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$
falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Definition

The operating ratio is

$$\Psi_s = \frac{\Phi}{C} S M_s. \quad (1)$$

The Proto-Regime refers to simulations where Φ is forced high while C is forced toward a numerical floor. In this limit, ordinary flow-through- constraint interpretation becomes unstable because the denominator approaches zero.

3 Why It Matters

A good runtime must know where its own equations stop behaving like ordinary domain models. The Proto-Regime is one such boundary. It marks a region where the model triggers warnings, finite-value checks, and explicit caveats.

4 Experiment Surface

The local script `experiments/discover_proto_regime.py` is the named entrypoint for probing this boundary. Related discovery machinery appears in:

- `experiments/notebooks/discovery_frontier_sweep.py`;
- `experiments/reports/NEW_DISCOVERIES_REPORT.md`;
- `experiments/reports/FRONTIER_DISCOVERIES.md`.

These reports are simulation logs. A high novelty score or new schema name means the engine found an unusual modeled state, not that nature has accepted the interpretation.

5 Mathematical Boundary

Let $C = \epsilon$ with $\epsilon > 0$ small. Then

$$\Psi_s \sim \frac{\Phi}{\epsilon} S M_s. \quad (2)$$

For fixed Φ , S , and M_s , Ψ_s diverges as $\epsilon \rightarrow 0$. The question is not whether the number becomes large; that is expected. The real question is whether the state produces reproducible structure under bounded regularization, or whether it is merely a division-by-small-number artifact.

6 Part I Boundary Marker

The active Part I boundary marker used here is the Paper XII superconductivity reallocation sample with transport index

$$\mathcal{S}_{\text{SC}} = 4.187183192107868$$

at the row where $T/T_c = 0.5287891440501045$. That row appears in the Part I Paper XII superconductivity reallocation table and belongs to the Part I transport-test program [5]. The runtime Proto-Regime is therefore written as a boundary family anchored to the Part-I-derived 4.187183192107868 transport marker.

7 Interpretation Rules

The Proto-Regime may be used as:

- a numerical stress test;
- a boundary marker for model validity;
- a candidate theoretical state for future mathematical analysis;
- a warning system for non-physical parameter choices.

It does not establish:

- frictionless travel;
- physical vacuum manipulation;
- a device claim;
- a regulated-use directive.

8 Falsification and Controls

Controls include:

1. increasing the constraint floor ϵ and checking whether the pattern vanishes;
2. adding finite S and M_s caps;
3. varying grid resolution and time step;
4. rerunning with different seeds and boundary conditions;
5. comparing against a known numerical singularity test.

9 Connection to the CLI

The CLI now needs a safe boundary-test command. Until then, Proto-Regime experiments remain Python scripts and report entries. Any future CDFL version needs an explicit warning when C is below a declared physical or domain-specific floor.

10 Conclusion

The Proto-Regime is important precisely because it is dangerous to overread. It marks a boundary where the runtime can teach its authors which claims need more mathematics, more controls, and stronger validation before they become physics.

References

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